

# /crew:autopilot

This guide is task-first. Each section answers one question you will have while autopilot drives a ticket: how to switch it on, how to hand it work, what its output means, and why it stopped. Every sample of script output below was printed by the script itself in a throwaway repository; the paths in them are shortened to `/path/to/repo`. Features that have not landed yet are listed once, under "What is coming", with their ticket ids and no invented output.

## What Autopilot Is

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- It drives **one ticket** through spec, plan, approval, implement, refresh, review, done and ship.
- It follows each phase's own command (`/crew:spec`, `/crew:plan`, `/crew:implement`, ...) in the same session, exactly as if you had typed it.
- It does not choose the order. Every turn it asks `crew_autopilot.py next`, which reads the ticket's files on disk and names the next phase, so a skipped phase is visible.
- It stops wherever a person is needed: a direction to approve, review findings with a BLOCK, an action on the never-without-a-yes list, or anything it cannot tell.
- It is **off by default**. Nothing happens until you arm it.

## Arm It

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Autopilot reads `.crew/config.json` in the repository. Only the exact string `plan` arms it:

```
{  
  
  "autopilot": { "mode": "plan", "maxPhases": 12 }  
  
}
```

`maxPhases` is how many phases one invocation may run before it stops and reports (default 12). Check what is in force:

```
python3 -B plugin/crew/hooks/scripts/crew_autopilot.py settings --root .
```

In a repository where `crew` is installed as a plugin, the script is under the plugin's `hooks/scripts/` directory; `/crew:autopilot` runs it for you. A fresh repository with only

"mode": "plan" set prints:

```
mode=plan maxPhases=12 deploy=none maxAutoReplans=0

approval=risk questions=risk

sleep=off schedule=none approval=- questions=- source=schedule
```

- **Line 1** is what autopilot may do: armed (mode=plan), at most 12 phases per run, no unattended deploy (deploy=none), no automatic replanning after a review runs out of rounds (maxAutoReplans=0).
- **Line 2** is who decides an approval and an open question: risk for both (see "Approvals").
- **Line 3** is the sleep schedule: off, so the day values on line 2 apply all the time.
- Any `warning:` line below them names a setting crew read as something else, and why.

## Hand It Work

### You type

```
/crew:autopilot
```

```
/crew:autopilot <id> OR /crew:autopilot run
<id>
```

```
/crew:autopilot status
```

### What happens

picks the ticket (below) and drives it

drives that ticket

reports where the ticket stands; runs nothing

**Which ticket, when you name none.** In this order: the handoff's `resume:` line (only when its `branch:` and `head:` match this checkout), then this worktree's active ticket, then `.work/INDEX.md` only when exactly one ticket is open. Autopilot prints where the ticket came from and every source it fell through. It never picks from a "next action" list.

**Work that has no ticket yet.** Write the request as a direction file under `.work/autopilot/` and mint a ticket from it:

```
python3 plugin/crew/hooks/scripts/crew_ticket.py assign --root . --direction-
file .work/autopilot/csv-export.md
```

`crew_ticket.py assign` refuses, minting nothing, when the file is not under `.work/autopilot/`, when autopilot is not armed, or when the direction is missing a section.

It writes the ticket as ready with the direction you gave; approval still goes through the policy below. The `/crew:autopilot assign` command that does this from plain text has not landed (see "What is coming").

## Read Its Output

### `/crew:autopilot Status`

status is read-only, armed or not. For a ticket whose direction is approved and which has no spec yet:

```
mode: plan, maxPhases 12

ticket: ABC-510 (from .work/INDEX.md)

phase: spec - next: /crew:spec ABC-510

waiting on: autopilot - run /crew:autopilot to continue

review: EMPTY, 2 of 2 rounds left

resume: no .work/HANDOFF.md

fell through: no .work/HANDOFF.md

fell through: no active-ticket entry for this worktree
```

| Line          | Meaning                                                                   |
|---------------|---------------------------------------------------------------------------|
| mode:         | armed or not, and the phase budget                                        |
| ticket:       | which ticket, and where it was found                                      |
| phase:        | the next phase and its command, or stopped with the reason                |
| waiting on:   | who has to act: autopilot (run it again), or owner (you) and what to type |
| review:       | the review ledger: no round yet, and how many rounds are left             |
| resume:       | what the handoff file says, if there is one                               |
| fell through: | each ticket source that did not answer, in order                          |

unknown anywhere means crew could not tell, never "nothing to do". With the plan written but not approved, the same ticket reads:

```
phase: approve, stopped - ABC-510 has no approved plan. The human types
/crew:approve ABC-510. status reads no approval policy: crew_autopilot.py settings
shows whether autopilot.approval lets /crew:autopilot approve it instead

waiting on: owner - types /crew:approve ABC-510
```

## The Next Line

Every turn autopilot runs `crew_autopilot.py next --root . --ticket <id>` and gets one line:

```
phase=spec stop=0 command=/crew:spec ABC-510 reason=no spec.md
```

- phase - the phase it names from disk.
- stop - 0: run command now. 1: stop and tell you reason and the command you type.
- command - the phase command, or empty when only you can act.
- reason - why, in words.

No output, a traceback or a non-zero exit is a stop, never a guess.

## "Autopilot Is Armed ... Its First Phase Is Spec"

When you start autopilot on a ticket whose direction you approved, the session reports something like this (it is the session's report, not a line a script prints):

```
Autopilot is armed (mode=plan, maxPhases 12) and has activated ABC-510. Its
first phase is spec: /crew:spec ABC-510
```

That is **not a stop**. With an approved direction and no `spec.md`, next answers `phase=spec stop=0`, so autopilot runs `/crew:spec ABC-510` straight away. The two phases where it stops before any work are `brainstorm` (no direction yet) and `direction-approval` (a direction that waits for your yes). For the same ticket while its INDEX row still says `direction`:

```
phase=direction-approval stop=1 command= reason=INDEX.md status is `direction`:
direction.md waits for the owner's yes in /crew:brainstorm
```

## Approvals

**One ticket.** Type `/crew:approve ABC-510`. The approval records a hash of `spec.md` and `plan.md`; any later edit to either, other than the header's status value, makes it **stale**, and autopilot stops at approve again until you re-approve.

**Several tickets at once.** `/crew:approve T-4 T-5`, or a range `/crew:approve T-0010..T-0012`. `crew` shows a PENDING list with each ticket's plan and spec hash and records nothing until you type `/crew:approve --confirm` as your next prompt.

**Letting autopilot approve.** `autopilot.approval` and `autopilot.questions` each take one of three policies:

| Policy            | Approval of a plan                                                      | An open question                       |
|-------------------|-------------------------------------------------------------------------|----------------------------------------|
| human             | always stops for you                                                    | always stops for you                   |
| self              | autopilot approves                                                      | autopilot takes the recommended option |
| risk<br>(default) | autopilot approves only a spec whose header says <code>risk: low</code> | the same rule, by the ticket's risk    |

A missing or unreadable risk counts as high. Every `autopilot.approval` setting also needs `scope.allowCliApproval: true` in `.crew/config.json`, which defaults to `false`, so out of the box nothing self-approves (`autopilot.questions` does not need it). Refused:

```
refused: scope.allowCliApproval is not exactly true in .crew/config.json, so no approval but the human's counts; the human types /crew:approve ABC-510
```

Allowed (`"approval": "self"` and `"scope": {"allowCliApproval": true}`):

```
self-approved ABC-510 under approval=self, risk=low

phase=implement stop=0 command=/crew:implement ABC-510 reason=approved, no review round under this plan
```

Autopilot reports every self-approved line by name in its final report. A ticket minted by `crew_ticket.py` assign follows the same policy as any other.

**Overnight.** `autopilot.sleep.schedule ("HH:MM-HH:MM", local time, default off)` names one nightly window in which `autopilot.sleep.approval` and `autopilot.sleep.questions` replace the two day values, so an overnight run can approve where the day setting would stop. `/crew:autopilot sleep` starts sleep mode now and `/crew:autopilot wake` ends it; both need

`scope.allowCliApproval: true`, and until L-1504 a manual sleep only ever tightens a policy. A plan approved asleep stops standing when the window ends if the day policy would not have approved it. Without the setting:

```
refused: scope.allowCliApproval is not exactly true in .crew/config.json, so only
the schedule puts autopilot to sleep
```

## What It Stops For

**Always a person.** brainstorm (no approved direction) and review-acceptance: review findings with any BLOCK, or a round the ledger's guarded accept refuses. A BLOCK is never accepted by autopilot, at any setting.

**Enforced by next from disk, every turn.** needs-replan, needs-replan-or-revert, unknown-ledger, failed-validate, direction-unknown, index-disagreement, unsettled-artifact, ticket-mismatch, max-phases, no-progress, auto-replan-cap and drift. `crew_autopilot.py stops --root .` lists them from the code.

**Never without an explicit yes.** Offboarding a role (`offboard-role`), deleting a code map or diagram (`delete-map`), rewriting the repository's crew metrics file (`rewrite-metrics`), and any force-push, branch delete, history rewrite or removal of a tracked file (`git-destruction`).

**Deploys.** `autopilot.deploy` is `none` (default), `nonprod` or `all`. Production needs `all` **and** `environments.prodUnattended: true` in both config layers, with the cloud guard in block. `crew_autopilot.py deploy-allowed` answers for one environment:

```
verdict=allow env=staging class=nonProd reason=autopilot.deploy=nonprod allows
nonProd

verdict=ask env=prod class=prod reason=autopilot.deploy is nonprod; production needs
all
```

Nothing in this crew version dispatches a deploy itself; settings warns you of that when the key is not none.

**Being told it stopped.** With crew's notifications configured (Telegram or Teams, in the machine-global config), a ping is sent when Claude Code stops and waits on you, led by a subject such as Question and naming the ticket and what it waits on; there is no ping per phase.

**Review.** A review phase ends at its verdict: autopilot reports the BLOCK and FIX lines verbatim and goes back to next. Fixing, accepting and replying on the PR are yours. One

exception: after the ledger says review: auto-accept: eligible on a final round with no BLOCK, autopilot runs review\_ledger.py --auto-accept, the guarded accept that refuses anything it cannot read cleanly.

**Ship.** After /crew:done, crew\_autopilot.py ship pushes the branch (never force), opens the PR, and under autopilot.ship: merge (the default; pr stops at the open PR) merges it with a merge commit only when every required check passes and the review receipt still stands. A high-risk ticket never merges on same-family reviews. It prints action=... stop=... pr=... reason=....

## Resume After /clear

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When context runs low, autopilot writes a handoff with a resume line:

```
resume: /crew:autopilot ABC-510
```

After /clear (or in a new session), type /crew:autopilot with no id: the handoff's resume: line names the ticket, but only while its branch: and head: match this checkout. A handoff from another branch or an older commit falls through to the active ticket, and status says so on a fell through: line. With auto-resume configured, crew can also type that resume command into its own session for you after the context cycle.

**Plain text.** With route.enabled: true in .crew/config.json (off by default), a whole prompt of continue, keep going or carry on routes to whatever next names for the ticket, and autopilot status or what is autopilot doing routes to /crew:autopilot status, and heading to bed or I'm back to /crew:autopilot sleep or wake. A continue whose next step is an approval asks instead: no phrase ever approves.

## Command Table

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| Command                                                               | Status | What it does                                        |
|-----------------------------------------------------------------------|--------|-----------------------------------------------------|
| /crew:autopilot,<br>/crew:autopilot <id>,<br>/crew:autopilot run <id> | landed | drive one ticket                                    |
| /crew:autopilot status                                                | landed | report, read-only                                   |
| /crew:autopilot focus<br><id>, focus, focus off                       | landed | lock autopilot onto one ticket, show it, release it |

| Command                                        | Status                                                                 | What it does                                                                          |
|------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| /crew:autopilot sleep,<br>/crew:autopilot wake | landed                                                                 | start or end sleep mode now (needs<br>scope.allowCliApproval: true)                   |
| /crew:autopilot assign                         | arrives with L-0611<br>(the router's<br>message still names<br>T-0019) | mint a ticket from plain text                                                         |
| /crew:autopilot goal                           | arrives with T-0012                                                    | work towards a goal across tickets                                                    |
| crew_autopilot.py<br>settings                  | landed                                                                 | the settings in force                                                                 |
| crew_autopilot.py status                       | landed                                                                 | the status lines above                                                                |
| crew_autopilot.py next                         | landed                                                                 | the one next line                                                                     |
| crew_autopilot.py resume                       | landed                                                                 | which ticket, and from where                                                          |
| crew_autopilot.py route                        | landed                                                                 | how the command's arguments are<br>read                                               |
| crew_autopilot.py approve                      | landed                                                                 | self-approval under the policy, or<br>refused:                                        |
| crew_autopilot.py<br>questions-check           | landed                                                                 | checks a researched answer before<br>it is taken                                      |
| crew_autopilot.py stops                        | landed                                                                 | every stop, from the code                                                             |
| crew_autopilot.py deploy-<br>allowed           | landed                                                                 | allow, ask or refuse for one<br>environment                                           |
| crew_autopilot.py ship                         | landed                                                                 | push, PR and guarded merge after<br>/crew:done                                        |
| crew_autopilot.py auto-<br>reject              | landed                                                                 | under maxAutoReplans, reject an out-<br>of-rounds BLOCK round for a<br>successor plan |
| crew_autopilot.py focus,<br>sleep, wake        | landed                                                                 | the writers behind the three<br>commands above                                        |

## What Is Coming

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Not landed in this crew version. No output is shown for any of them; each ticket adds its own section here when it lands.

| Example                                                           | Ticket                 | Phrase      |
|-------------------------------------------------------------------|------------------------|-------------|
| Approving a goal's split, goal:<slug>                             | T-0012                 | goal:<slug> |
| /crew:autopilot assign, from plain text                           | L-0611                 | assign      |
| A goal across several tickets                                     | T-0012, L-0541         | goal        |
| A wave: tickets run in parallel                                   | T-0029                 | wave        |
| Autopilot's own split into PR slices                              | T-0052, T-0058, T-0059 | split       |
| A goal that survives /clear, a branch switch and a crash (--goal) | T-0056                 | --goal      |
| The full goal walkthrough                                         | T-0012, L-0541         | walkthrough |

## When Something Looks Wrong

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- **"It stopped at spec, but I wanted it to keep going."** It did not stop there: phase=spec stop=0 runs. Read the next line; a real stop has stop=1 and a reason.
- **"It will not approve my plan."** scope.allowCliApproval is not true, or the policy is human, or the ticket's risk is above what risk allows. The refused: line says which.
- **"It keeps stopping at approve after I edited the spec."** The approval went stale. Re-approve.
- **"It stopped at review-acceptance."** A BLOCK, or a round the guarded accept refused. That is yours to fix or accept.
- **"unknown in status."** crew could not read something it needed; the same line says what.
- **Stale installs, hook noise, review loops and turning things off** are in the troubleshooting guide, "crew - Troubleshooting".